

ILLINOIS TECH | College of Computing

ITMM 485 / 585
Dr. Gurram Gopal

Legal and Ethical Issues in
Information Technology



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Ethical Concepts and Theories

ILLINOIS TECH | College of Computing

Ch8: Intellectual Property Issues
P2: Copyright and Fair Use



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Learning Objectives:

Upon completion of this lesson the students should be able to:

- Explain and discuss the philosophical and legal arguments used to protect intellectual property
- Describe the differences and similarities between the Free Software Foundation & the Open Source Initiative
- Explain how Creative Commons licensing of IP works
- Discuss the balance between public good and profit for IP rights holders

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Module Objectives:

Upon completion of this lesson the students should be able to:

- Define intellectual property (IP) and related terms including copyright
- Explain fair use and how it applies to digital data and IP

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Copyright Protection

- Copyright law in the Anglo-American world was enacted in response to concerns about certain uses of printing-press technology.
- Two different kinds of concerns arose in response to the widespread publishing of pamphlets made possible by the printing press:
 - a) the British monarchy wanted to control the spread of "subversive" and "heretical" works being printed; and
 - b) authors wanted to protect their creative works from being reproduced without their permission.

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Copyright Protection (Continued)

- The English Statute of Anne (1710) was the first law to give protection to authors for works attributed to them.
- The American Colonies followed English law regarding copyright.

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Copyright Protection (Continued)

- In the US Constitution, a specific provision in Article 1, Section 8 states:
The congress shall have the power... to promote the Progress of Science and the useful Arts, by securing for limited Times to authors and inventors the exclusive Rights to their respective Writings and Discoveries.

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Evolution of U.S. Copyright Law

- The first copyright law was enacted in 1790.
- It applied primarily to books, maps, and charts.
- The law was later extended to include newer forms of media such as photography, movies, audio recordings, and so forth.
- In 1909, the copyright law was amended to include any "form that could be seen and read visually" by humans.
- This change was in response to a new technology: the player piano.

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Evolution of Copyright Law in the U.S. (Continued)

- The 1909 change was prompted by a case in 1908 involving a song that was copied onto a perforated piano music roll.
- Since the musical copy could not be read visually (by humans) from the piano roll, the copy was not considered a violation of the song's copyright.
- The "machine readable" vs. "human readable" distinction had implications for decisions about whether software programs could qualify for copyright protection.

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Evolution of Copyright Law in the U.S. (Continued)

- A software program's source code can be read by humans.
- Its "executable code," which "runs" on a computer, cannot be read by humans.
- Beginning in the 1960s, arguments were made that computer programs should be eligible for copyright protection.

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Evolution of Copyright Law in the U.S. (Continued)

- Copyright law was significantly modified again in 1976.
- Under the 1976 Copyright Act, computer programs still did not clearly satisfy the requirements necessary for making them eligible for copyright protection.
- The Copyright Act was amended again in 1980 to address the status of software programs.

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Evolution of Copyright Law in the U.S. (Continued)

- In 1976, the concept of a literary work was extended to include:
- programs,
- computers,
- databases that "exhibit authorship."

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Evolution of Copyright Law in the U.S. (Continued)

- A computer program was defined under the US Copyright Act as
a set of statements or instructions to be used directly in a computer in order to bring about certain results.
- To get a copyright for a computer program, the author had to show that the program contained an *original expression of ideas* and not simply the ideas themselves

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Evolution of Copyright Law in the U.S. (Continued)

- In 1998, two important amendments were made to the 1976 Copyright Act:
 - 1) Sonny Bono Copyright Term Extension Act (SBCTEA),
 - 2) Digital Millennium Copyright Act (DMCA).
- The SBCTEA extended the length of copyright protection from the life of the author plus 50 years to the life of the author plus 70 years.

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Evolution of Copyright Law in the U.S. (Continued)

- Protection for "works of hire" (often commissioned by corporations) produced before 1978 were extended from 75 years to 95 years.
- Critics of the SBCTEA noted that the law was passed just in time to keep Mickey Mouse from entering the public domain.
- Critics also pointed out that the Disney Corp. lobbied very hard for the passage of this act.

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Evolution of Copyright Law in the U.S. (Continued)

- The DMCA has also been severely criticized, because of the manner in which the rights are extended.
- The DMCA potentially restricts the development and use of digital technology.
- It contains a controversial "anti-circumvention clause," which forbids:
the development of any software or hardware technology that *circumvents* (or devises a technological workaround) to copyrighted digital media.

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What Does Copyright Law Protect?

- A copyright is a legal form of protection given to a "person" or author.
- The author can be an entity such as organization or a corporation, such as Microsoft, as well as an individual.
- A copyright protection is given for the expression of an idea such as a book, poem, musical composition, photograph, dance movement, motion pictures, audiovisual works, or computer software.

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Copyright Protection

- For a work to be protected under copyright law, it must satisfy three conditions in that it needs to be:
 - 1) original;
 - 2) non-functional;
 - 3) fixed in a tangible medium.

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Copyright Protection (Continued)

- Copyright holders have the exclusive right to:
 - make copies of the work;
 - produce derivative works, translations into other languages, movies based on the book, and so forth;
 - distribute copies;
 - perform works in public (musicals, plays. etc.);
 - display works in public (e.g., art works).

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The *Fair Use* Provision in Copyright Law

- The principle of *fair use* balances the exclusive controls given to copyright holders against the broader interests of society.
- Fair use means that an author or publisher may make limited use of another person's copyrighted work for purposes such as:
 - criticism,
 - comment,
 - teaching,
 - scholarship,
 - research.

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Fair Use (Continued)

- The fair-use principle has also supported the practice of "reverse engineering."
- Reverse engineering is very important in the computer industry in particular, and in engineering in general, because it allows someone to buy a product for the purpose of taking it apart to see how it works.

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The *First-Sale* Doctrine in Copyright Law

- The first-sale doctrine is another balancing scheme in copyright law.
- It applies once the original work has been sold for the first time, when the original owner loses rights over the work of art.
- Once you purchase a copy of a book, audio tape, painting, etc., you are free to give away, resell, or even destroy the copy of that work.
- It is not clear whether one is permitted to give away digital versions of these works.

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SBCTEA: The Extension of Copyright Protection

- SBCTEA has been controversial (as noted above).
- See the discussion (in the textbook) involving Eric Eldred's Web site, which was set up to make available electronic versions of out-of-print books (in the public domain) that were then very difficult to locate.
- When SBCTEA passed, many books that were about to enter the public domain could not be included on Eldred's site.

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DMCA: The Extension of Copyright Protection

- The DMCA has a controversial anti-circumvention clause (described earlier).
- See the discussion (in the textbook) involving Dimitri Sklyarov, a graduate student who wrote a program to decrypt the code for an electronic book reader that was developed by Adobe (a U.S.-based software company).
- Adobe feared that with Sklyarov's program, "pirates" could read e-books for free.
- Sklyarov, a Russian citizen, was arrested under the DMCA's anti-circumvention clause, when he attended a conference in the U.S.

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The (Now Classic) LaMacchia Case

- In 1994, Robert LaMacchia, a student at MIT, was arrested for operating an electronic bulletin board system (BBS) system (*Cynosure*), on which users could upload/download proprietary (copyrighted) software.
- LaMacchia did not charge users for his service and he did not make any money from the Cynosure BBS; so, did he commit a crime?
- How did the incident involving LaMacchia and Cynosure foreshadow the Napster case?

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The Original Napster Case

- In 1999, the Napster Web site was sued by the Recording Industry Association of America (RIAA) for distributing copyrighted music on the Internet.
- Gnutella, Morpheus, and KaZaA have also distributed copyrighted music on the Internet.
- Unlike Napster, which used a centralized server, index, and registry of names, the other services used a “decentralized” system.

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The *Verizon v. RIAA* Controversy

- The Recording Industry took a different approach in challenging Gnutella, Morpheus, and KaZaA.
- It issued court subpoenas to Internet Service Providers (ISPs) such as Comcast and Verizon, as well as to major universities, for the names of users it suspected to have downloaded and exchanged large volumes of copyrighted music via those ISPs.
- In the *Verizon v. RIAA* case, Verizon challenged the RIAA in court on the grounds that complying with such requests violated specific articles of the U.S. Constitution.
- As a result of the RIAA’s suits, some universities have disabled their P2P file-sharing systems altogether.

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(Online) Music Streaming Services

- Review Scenario 8-1 in the textbook.
- In some ways, streaming services pose the next challenge for copyrighted music.
- Some critics of streaming services believe that these online services both:
 - a) contribute to piracy,
 - b) are responsible for the decline of CD sales.
- Streaming’s defenders offer a different view.

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Streaming Online Music (Continued)

- Spotify, a very popular streaming service, has argued that it provides members with an alternative to piracy (while, at the same time, providing revenue to recording artists that they otherwise would not receive).
- Spotify also points out that its members reported (in a survey conducted by Spotify) that they had paid either very little or nothing for music before joining that streaming service.
- So, Spotify claims that if the sale of CDs has fallen, it is not because of its service; thus, it cannot be held responsible for any decline in CD sales.

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Streaming Online Music (Continued)

- Spotify has also pointed out that it has explicit contractual agreements with the recording artists and record labels featured on its service.
- So, streaming services like Spotify are very different from the various P2P music sharing sites in the past, which enabled users to download unauthorized copies of proprietary recordings in violation of copyright law.

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Streaming Online Music (Continued)

- While many recording artists have agreed to Spotify's terms, others have refused to grant Spotify the right to have their songs played.
- Others have subsequently changed their mind and had their music removed; e.g., Taylor Swift, who did not allow her album *1989* to be aired on Spotify, has since removed all of her earlier music from that streaming service as well.
- Swift believes that Spotify's policies are "unfair" to recording artists because they significantly decrease the amount of income artists would receive if Spotify's users instead purchased music in the form of CDs and MP3s.

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Streaming Online Music (Continued)

- Are Spotify's practices unfair, as Swift claims, and for the reasons she suggests?
- First, we should note that Spotify is a service whose practices are in strict compliance with copyright law.
- So unlike "pirate" music sites, beginning with Napster and evolving to the present day, Spotify has legal contracts with recording artists and record labels; these contracts also include a royalty structure to which both parties agree.

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Streaming Online Music (Continued)

- But Swift is correct in claiming that recording artists – at least some of them – stand to gain far more royalty income by "going the song-for-purchase route" than from the royalties offered by Spotify.
- For example, recording artists who are currently in their prime, such as Swift and Beyoncé, may be significantly affected by the amount of royalty income they can potentially lose.

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Streaming Online Music (Continued)

- Not all, or even most, recording artists, are negatively affected in the same way as Swift apparently is by Spotify's policies, as well as those of other major streaming services.
- As some critics point out, some solo artists and bands who may be starting out in their careers can gain more exposure through services like Spotify than they otherwise would.
- Also, recording artists who are past the primes of their careers can benefit from still having their songs aired to Spotify users.
- So, Swift's claim that Spotify is "unfair" to recording artists does not seem to be completely accurate, since many artists do indeed stand to gain from exposure to their music via this online service.

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Streaming Online Music (Continued)

- We can now see how the debate over access to digital music has moved well beyond the earlier Napster-era questions pertaining to illegal downloading.
- For example, that debate now includes questions about legitimate modes of freely accessing/listening to copyrighted music via a process that financially rewards musicians and the recording industry.
- Of course, this debate is no longer limited to online music; it now includes issues affecting unauthorized access to (copyrighted) online movies as well.

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The *MGM v. Grokster* Case

- Review the discussion of the *MGM v. Grokster* case (in the textbook).
- MGM and several motion picture studios sued Grokster for "contributory copyright infringement" through its file-sharing service.
- MGM alleged that over 90% of the material exchanged on Grokster was copyrighted material and that the P2P service was legally liable for the material being exchanged.

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MGM v. Grokster (Continued)

- A district court disagreed with MGM, ruling that Grokster could not be held liable for the distribution of copyrighted material because:
 - 1) it lacked sufficient knowledge of the infringement, and
 - 2) it did not materially contribute to the copyright infringement.
- The case was later appealed to U.S. Supreme Court, which sided with MGM.

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Who owns the copyright to this picture?



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Copyright on Works Not Created by Humans

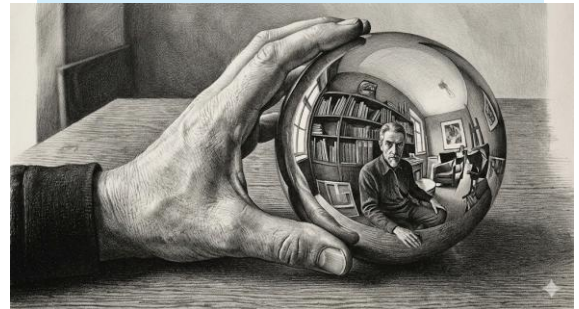
Monkey selfie copyright dispute

Disputes about the copyright status of selfie photographs taken by Celebes crested macaques, using equipment belonging to British wildlife photographer David J. Slater. In December 2014, the United States Copyright Office stated that works that lack human authorship, such as "a photograph taken by a monkey", cannot have their copyright registered at the US Copyright Office



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Are images/videos created by Dall-E, Nano Banana, etc. copyrightable?



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Are images/videos created by Dall-E, Nano Banana, etc. copyrightable?

In the United States, the Copyright Office has repeatedly stated that

- copyright requires human authorship.
- Works generated autonomously by AI don't qualify. In Thaler v. Perlmutter (2023), a federal court upheld this position, ruling that an image created entirely by AI without human creative input cannot be copyrighted.
- The Copyright Office has issued guidance saying it will not register works produced by AI without human creative control.

U.S. Supreme Court declines to hear dispute over copyrights for AI-generated material

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